NAME:-Sheikh Wasimuddin

ROLL NO:-73

BRANCH :-CSE(IOT)

LAB 12	
Aim:	Analyzing the Economic Impact of Vocational Training on Household Incomes • Dataset: India Human Development Survey (IHDS) • Key R functions/packages: o lm(income ~ vocational training + other factors) o car::vif() for multicollinearity check o summary() for interpreting regression outputs o ggplot2::geom_point() for scatter plots
Theory	❖ **Theory Behind Economic Impact of Vocational Training on Household Incomes** ❖ Vocational training is a critical factor in enhancing economic opportunities by equipping individuals with industry-specific skills. This structured learning approach focuses on practical applications rather than traditional academic knowledge, directly influencing employability and earnings. ❖ **1 Human Capital Theory** ❖ The **Human Capital Theory** suggests that education and skill development contribute to higher productivity, leading to increased earnings. Vocational training improves technical expertise, efficiency, and adaptability, making workers more valuable in the labor market. ❖ **2 Income Determination Model** ❖ Household incomes are influenced by multiple factors, including: ❖ **Education**: Higher levels typically lead to better-paying jobs. ❖ **Work Experience**: Skills acquired over time enhance earning potential. ❖ **Employment Sector**: Specialized industries often offer higher wages. ❖ **Vocational Training**: Provides job-ready skills, reducing unemployment risks and improving earnings. ❖ **3 Role of Vocational Training in Economic Mobility** ❖ Vocational training: ❖ Increases **earning potential** through skill enhancement. ❖ Reduces **income inequality** by providing opportunities to disadvantaged groups. ❖ Promotes **job stability**, helping individuals secure long-term employment. ❖ Improves **workplace productivity**, benefiting businesses and national economies. ❖ **4 Statistical Analysis in Economics** ❖ Economists use regression models to quantify the impact of vocational training on household incomes: ❖ **Linear regression** identifies relationships between vocational training and income. ❖ **Multicollinearity tests** ensure predictors are independent. ❖ **Data visualization** (scatter plots, histograms) helps detect trends in income distribution. ❖ **Conclusion** ❖ Vocational training is a powerful tool for economic growth, particularly in developing nations. By equipping individuals with practical, job-oriented skills, it

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	enhances household incomes, reduces unemployment, and strengthens overall economic resilience.

Program:	#SHEIKH WASIMUDDIN 23071044 CSEIOT-73
	
	# Load necessary libraries
	
	library(car)
	
	library(ggplot2)
	
	library(dplyr) # for data manipulation
	
	library(ggthemes) # for nice themes
	
	library(viridis) # for color palettes
	
	# Read the CSV file
	
	data <- read.csv("C:/Users/student/Downloads/ihds_vocational_training_sample.csv")
	
	# View the first few rows
	
	head(data)
	
	# Fit a linear regression model
	
	model <- lm(income ~ vocational_training + education_years + age + rural + household_size, data = data)
	
	# Summary of the model
	
	summary(model)
	
	# Check multicollinearity using Variance Inflation Factor (VIF)
	
	vif(model)
	
	# 1. Scatter plot: Income vs Education Years, colored by Vocational Training
	
	ggplot(data, aes(x = education_years, y = income, color = factor(vocational_training))) +
	
	  geom_point(size = 3, alpha = 0.7) +
	
	  scale_color_viridis(discrete = TRUE, option = "D") +
	
	  labs(

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```
    title = "Household Income vs Education Years",
    x = "Years of Education",
    y = "Household Income",
    color = "Vocational Training"
  ) +
  theme_minimal()

# 2. Box plot: Income by Vocational Training Status
ggplot(data, aes(x = factor(vocational_training), y = income, fill =
factor(vocational_training))) +
  geom_boxplot(alpha = 0.8) +
  scale_fill_viridis(discrete = TRUE, option = "E") +
  labs(
    title = "Income Distribution by Vocational Training Status",
    x = "Vocational Training (0 = No, 1 = Yes)",
    y = "Household Income"
  ) +
  theme_economist()

# 3. Density plot: Income distribution
ggplot(data, aes(x = income, fill = factor(vocational_training))) +
  geom_density(alpha = 0.6) +
  scale_fill_viridis(discrete = TRUE, option = "A") +
  labs(
    title = "Density Plot of Household Income",
    x = "Household Income",
```

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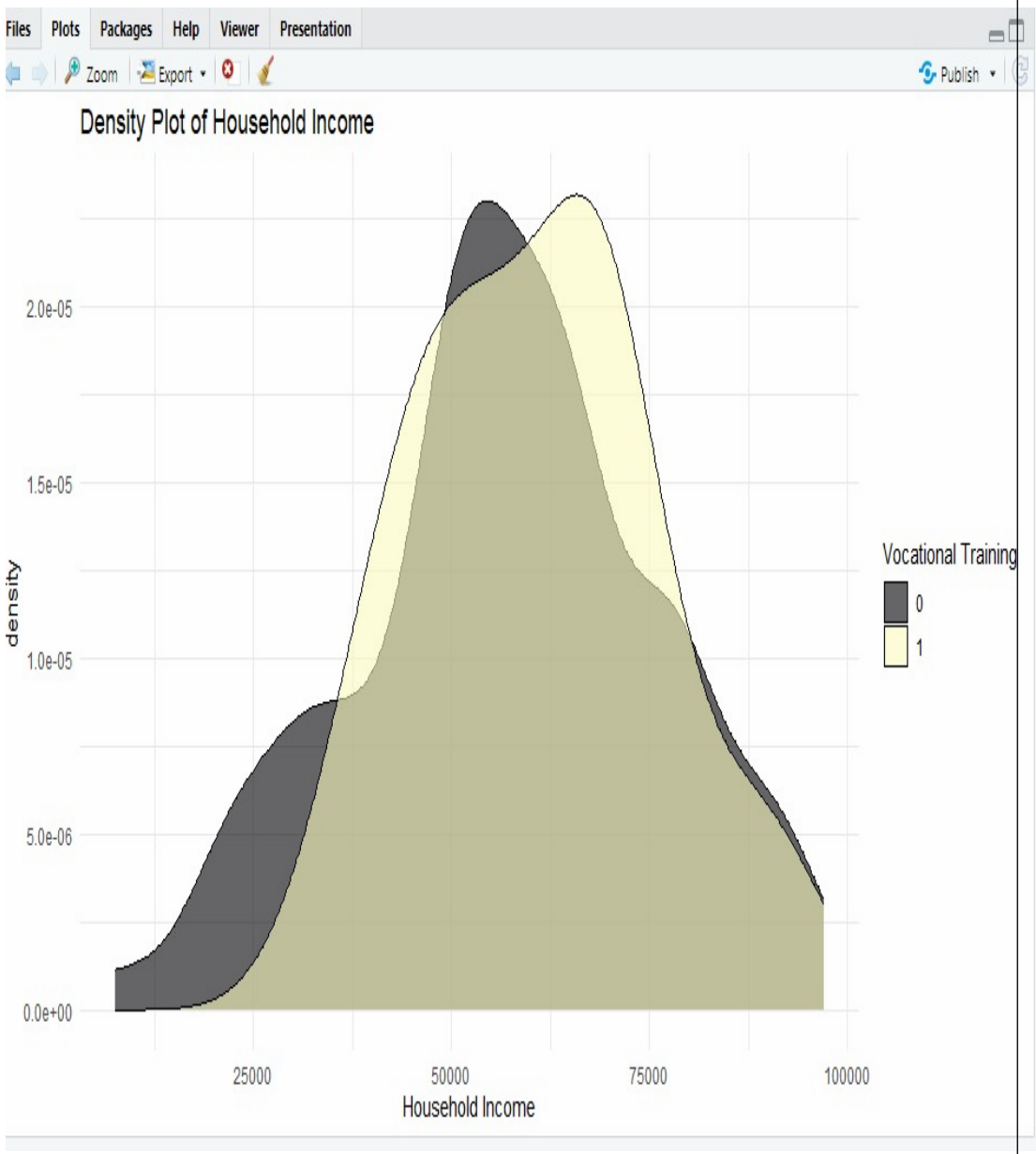
```
  fill = "Vocational Training"
) +
theme_minimal()
```

Output:



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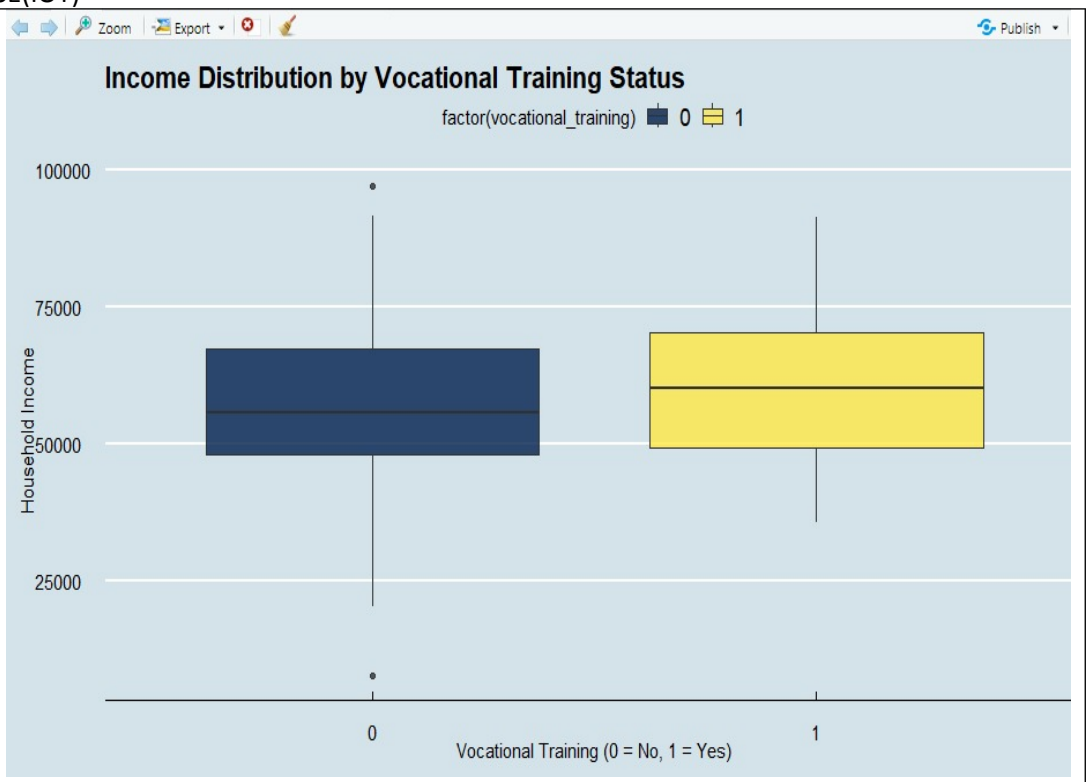

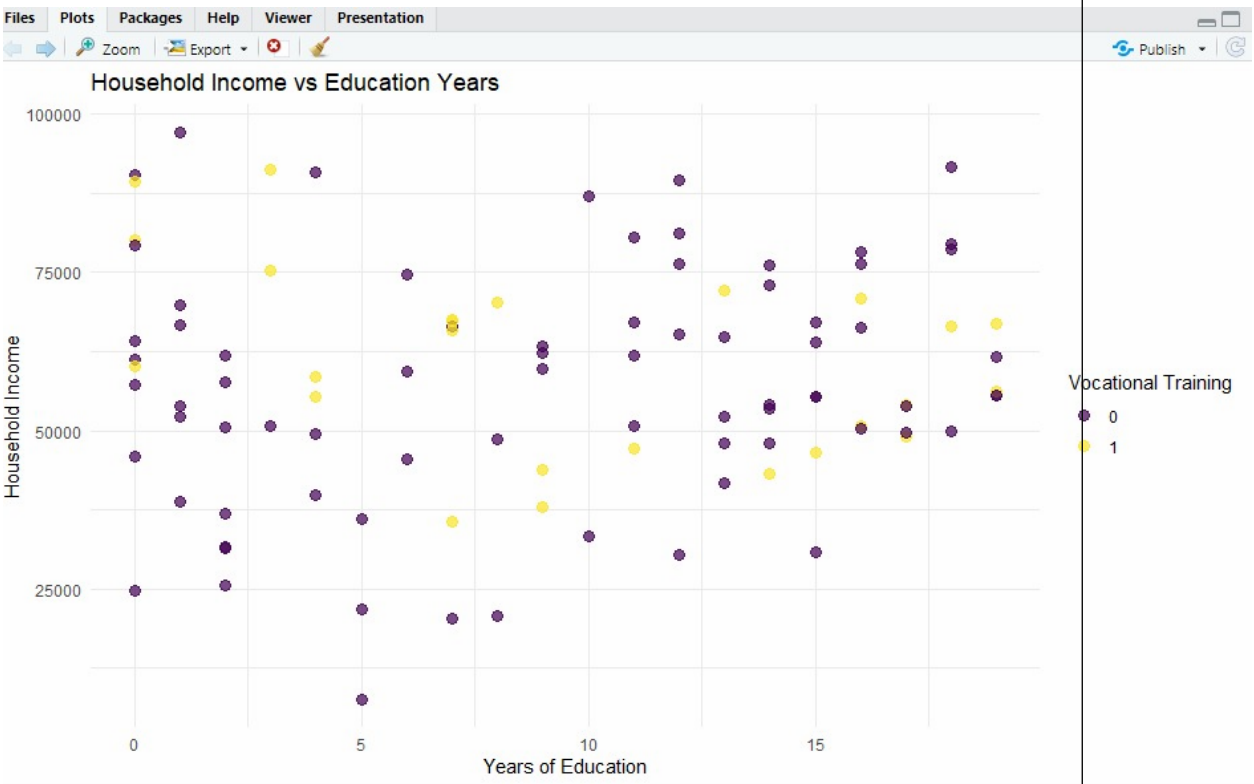

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Conclusion	Practical is executed successfully.
Name Of Faculty:	Prof. Akhil Jajulwar